Task 3 Methodology: Dual-Track Factor Attribution Analysis

INPUT

Observational Data
(Judge Scores)

Latent Variable Inversion
(Reconstructed Fan Votes)

Metadata Integration
(Age, Industry, Partner, State)

MODEL

Dual-Track Regression Model Framework
Model J: $Y_{Judge} = \beta^{(J)}X + \gamma^{(J)}Z + \varepsilon_J$
Model F: $Y_{Fan} = \beta^{(F)}X + \gamma^{(F)}Z + \varepsilon_F$

STATS

Welch's T-Test
Coefficient Structural
Divergence Testing

Intraclass Correlation (ICC)
Partner Effect
Quantification

Effect Size Indices
(η^2, Cohen's d, R^2)
Intensity Mapping

FINDINGS

Structural Split
(Age Sensitivity Gap)
Judge vs Fan Divergence

Partner Effect Paradox
(ICC Gap: 16% vs 5%)
Tech vs Fame Driver

Geographic Identity
(Regional Bias)
$\eta^2 = 9.5\%$